# Assignment: TravelNest Platform

## Business Case

TravelNest is a European online travel marketplace connecting travellers with accommodation providers across 10 countries.

Management has the following strategic question:

> *"What drives booking revenue and customer lifetime value, and how do accommodation quality, cancellation behaviour, and geography impact long-term platform performance?"*

Your task is to build a data pipeline and answer this question with evidence.

## Technical Requirements

You must implement a **Medallion Architecture**:

- /Raw – original source files, never modified
- /Bronze – raw ingestion, no transformations
- /Silver – cleaned and standardised
- /Gold – dimensional star schema, business-ready

**Technology:** Google Colab · PySpark · Spark SQL · Matplotlib / Seaborn

## Source Files

The following files are provided. **You may not assume the data is clean.**

| File | Description |
|---|---|
| bookings.csv | All booking transactions |
| customers.csv | Customer profiles |
| accommodations.csv | Accommodation properties |
| rooms.csv | Room types per accommodation with pricing |
| payments.csv | Payment records per booking |
| reviews.csv | Customer reviews per booking |
| accommodation_categories.csv | Reference table for accommodation categories |

You may justify excluding a file if properly explained.

## Bronze Layer (Raw Ingestion)

**Requirements:**

- Load all CSVs as-is — no schema modifications, no transformations
- Store as Parquet
- Add ingestion timestamp (audit metadata only)

**Show per table:**

- Schema
- Row count
- Sample records

## Silver Layer (Cleansed & Standardised)

### ✓ Cleaning Requirements

The data comes directly from operational systems and has not been validated. Analyse each source file, document every data quality issue you find, and justify how you handled it.

**Your pipeline must be robust enough to handle unexpected data gracefully. For every decision you made, document what you encountered and why you handled it that way.**

### ✓ Implement SCD Type 2

Implement SCD Type 2 for dimensions where historical changes are relevant.

Each SCD2 table must include:

- surrogate_key
- valid_from
- valid_to
- is_current flag

### ✓ Keys

- Generate surrogate keys for all dimensions
- Keep business keys

## Gold Layer (Business Model – Star Schema)

Model a dimensional star schema that supports answering the strategic question.

### Required Fact Table

**fact_bookings**

Must include at minimum:

· booking_id · room_key · customer_key · accommodation_key · date_key · revenue · nights · booking_status · cancellation_flag · lead_time_days · review_score

You determine which dimensions are needed and justify your choices.

### Upsert Requirement

Gold tables must be written using MERGE (Upsert) logic. No full overwrite allowed.

## Advanced Analytical Requirements

### ✓ Cancellation KPI

Define and compute a cancellation performance metric. Classify bookings as:

- **On-time cancellation** — cancelled more than 7 days before check-in
- **Late cancellation** — cancelled 7 days or fewer before check-in
- **No-show** — booking status is no_show
- **Completed** — booking was fulfilled

Justify your classification logic.

### ✓ Lead Time Analysis

Compute **lead_time_days** = check-in date – booking date.

Use this to analyse whether customers who book far in advance behave differently from last-minute bookers (revenue, review score, cancellation rate).

### ✓ Use at least 2 Window Functions

Examples:

- RANK()
- DENSE_RANK()
- SUM() OVER()
- ROW_NUMBER()

## Required Business Insights (7 Mandatory Visualisations)

| # | Visualisation |
|---|---|
| 1 | Revenue by accommodation type (all types) |
| 2 | Top 10 customers by lifetime revenue (use window functions) |
| 3 | Monthly revenue trend (time series) |
| 4 | Cancellation rate (%) by country |
| 5 | Top 10 cities by total revenue |
| 6 | Revenue vs. average review rating per accommodation (correlation) |
| 7 | Customer lifetime value distribution (Low / Medium / High using CASE) |

For each visualisation you must:

- Show the chart
- Write 1 sentence explaining why you chose this specific visualisation

• Write 2–3 sentences on what you observe and what it means for the business — use specific numbers from your output

## Deliverables

### One Colab Notebook (.ipynb)

Must contain:

• Clear comments per layer
• Explanation of all modelling decisions
• Documentation of data issues encountered and how the pipeline handles them
• SCD logic explanation
• 7 visualisations with written interpretation

### PowerPoint Presentation

Must include:

1. Business understanding
2. Architecture design (Bronze / Silver / Gold)
3. Star schema diagram
4. SCD explanation
5. Pipeline decisions — what data issues did you encounter and how did you handle them?
6. 7 visualisations with interpretation
7. Final answer to the strategic management question — supported by your data